

Stock Market Trend Analysis

Detailed Algorithm Architecture & Dataset Specification

SECTION 1: ALGORITHM SPECIFICATION

1.1 Model & Technique Applied

The system utilizes a **Hybrid Deep Learning & NLP Model Architecture** combining:

- **Stacked Long Short-Term Memory Network (LSTM):** Captures long-term dynamic time-series dependencies in quantitative price data without vanishing gradients.
- **Gated Recurrent Unit (GRU) Layer:** Accelerates convergence and models short-term market volatility dependencies.
- **FinBERT (Financial Bidirectional Encoder Representations from Transformers):** Natural Language Processing transformer fine-tuned on financial corpora to extract market sentiment scores ($S_t \in [-1, 1]$) from news headlines.
- **Dense Feed-Forward Fusion Network:** Merges LSTM/GRU temporal latent representations with FinBERT sentiment embeddings to output binary trend signals and continuous price estimates.

1.2 Logic and Method Used

The fundamental logic relies on combining quantitative price velocity with qualitative market sentiment. Financial time-series data $X_t = [Price_t, Vol_t, RSI_t, MACD_t, S_t]$ is structured into rolling 60-day window sequences. The system predicts future directional movement using a hybrid loss function that balances Mean Squared Error (MSE) for price levels and Binary Cross-Entropy (BCE) for trend direction ($Up = 1, Down = 0$).

1.3 Step-by-Step Computational Procedure

Step 1: Data Ingestion & Alignment

Extract historical OHLCV data alongside news feed timestamps. Align unstructured news headlines with daily market trading dates.

Step 2: Technical & Sentiment Feature Calculation

Compute technical indicators (\$EMA\$, \$RSI\$, \$MACD\$). Compute sentiment polarity scores \$S_t\$ using FinBERT NLP transformers.

Step 3: Sliding Window Sequence Generation

Transform continuous feature matrices into rolling 3D sequence tensors of dimension \$(N_{\text{samples}}, 60_{\text{timesteps}}, M_{\text{features}})\$.

Step 4: Deep Learning Inference & Forward Pass

Pass sequence data through stacked LSTM layers (128 & 64 units) and a GRU layer (32 units). Combine outputs with Dense linear layers.

Step 5: Output Post-Processing & Decision Signal Generation

Apply sigmoid activation for direction probability (\$P_{\text{up}}\$). If \$P_{\text{up}} > 0.55 \rightarrow\$ Bullish Alert; If \$P_{\text{up}} < 0.45 \rightarrow\$ Bearish Alert; Else \$ \rightarrow\$ Neutral/Hold.

1.4 System Process Flow

The operational data pipeline follows a sequential, closed-loop processing flow:

- Ingestion:** APIs fetch daily price series and raw financial headlines.
- Preprocessing & Normalization:** Feature scaling via Min-Max Scaler (\$[0,1]\$ bounds) and NLP text cleaning.
- Feature Matrix Fusion:** Merging technical time-series matrices with sentiment vectors.
- Model Inference:** Processing vector windows through the LSTM-GRU architecture.
- Visualization & Display:** Rendering output visual charts and trend indicators on an interactive dashboard interface.

SECTION 2: DATASET SPECIFICATION

2.1 Data Sources & Metadata

- **Historical Price Data Source:** Yahoo Finance (‘yfinance API’) and Alpha Vantage API.
- **Financial News Data Source:** NewsAPI and Financial News Scraping feeds (Bloomberg/Reuters aggregated headlines).
- **Data Format & File Type:** Time-indexed CSV files and JSON API payload response streams.
- **Data Volume / Sample Size:** 10+ years of daily historical data (~2,520 trading days per equity ticker) spanning across major equity indices (e.g., S&P 500, NIFTY 50, NASDAQ Tickers).

2.2 Dataset Schema (Features & Attributes)

Attribute Name	Data Type	Description & Mathematical Formula
‘Date’	DateTime (YYYY-MM-DD)	Primary key index representing the trading day session.
‘Open’, ‘High’, ‘Low’, ‘Close’	Float64	Session opening, highest, lowest, and daily closing prices (USD/INR).
‘Adj Close’, ‘Volume’	Float64 / Int64	Adjusted closing stock price and total volume of shares traded.
‘RSI_14’	Float64	Relative Strength Index computed over a 14-day lookback window.
‘EMA_20’, ‘EMA_50’	Float64	Exponential Moving Averages for short-term and medium-term trends.
‘MACD’ & ‘Signal_Line’	Float64	Moving Average Convergence Divergence difference and signal vector.
‘Sentiment_Score’ (\$S_t\$)	Float64 (\$[-1, 1]\$)	Daily aggregated headline news sentiment score derived via FinBERT.
‘Target_Direction’	Binary Int (0 or 1)	Ground truth target variable: \$1\$ if \$Close_{t+1} > Close_t\$, else \$0\$.

2.3 Dataset Partitioning (Train/Validation/Test Split)

To prevent target leakage inherent to random shuffling in financial time-series forecasting, chronological split partitioning is applied:

Dataset Split	Percentage Allocation	Usage & Time Horizon
Training Set	70%	Historical data (Years 1 to 7) used for network weights optimization.
Validation Set	15%	Intermediate data (Years 7 to 8.5) used for hyperparameter tuning & early stopping.
Testing Set	15%	Out-of-sample data (Years 8.5 to 10) for unbiased model evaluation.